

MTH 310/520 Graph Theory: Midterm

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Total: 30 points

Attempt all questions from Wall Maria. Students enrolled at 3xx level may choose to attempt all four questions from Wall Rose, or two from Wall Rose and one from Wall Sina, or both questions from Wall Sina. Students enrolled at 5xx level should either attempt any two from Wall Rose and any one from Wall Sina, or both questions from Wall Sina.

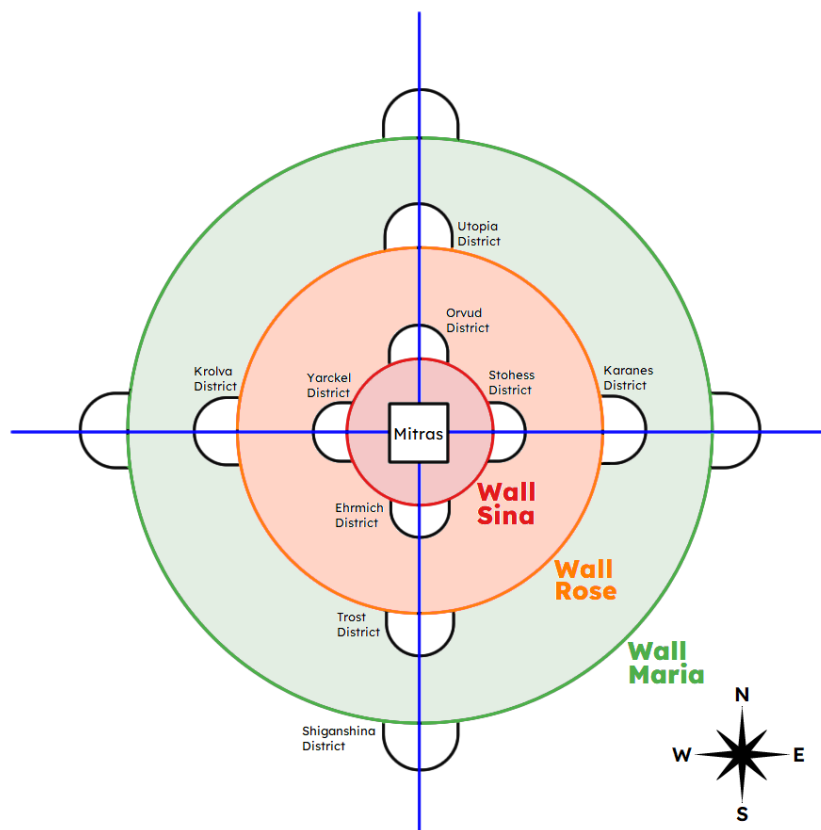


Fig. 1: Schematic of the three walls of Shingeki no Kyoujin, which are used to name the difficulty levels of this paper.

I. WALL MARIA

I.1 Is the sequence $(4, 2, 2, 2, 2)$ graphic? [CO1]

I.2 Prove or disprove: Every set of $\kappa(G)$ independent $u - v$ paths can be extended to (i.e., is contained in) a set of $\lambda(G)$ edge-disjoint $u - v$ paths. [CO3]

I.3 Prove or disprove: A k -regular graph is k -connected. [CO2, CO3]

I.4 Prove that for any $k \geq 1$, and for any $\{u, v\} \subseteq V(G)$, $d_{G^k}(u, v) = \lceil \frac{d_G(u, v)}{k} \rceil$. [CO2]

I.5 Can there be two (non-trivial) trees T_1 and T_2 with $|T_1| = |T_2|$ such that $T_1 \square T_2$ is a tree? [CO1]

[1+3+2+2+2=10 points]

II. WALL ROSE

II.1 Suppose you are given a set N of $2n$ real numbers. We say that a pair $\{u, v\} \subseteq N$ is *related* if $|u - v| \in (1, 2)$.

- Does there exist a triple $\{u, v, w\} \subseteq N$ such that u, v, w are pairwise related?
- Give an upper bound on the number of related pairs.

[Hint: You may look up Tutorial 2 questions and use any result from it.]

- For every n , give a set of $2n$ real numbers that satisfy the bound in II.1.b.

[Hint: Go back to the relevant problem of Tutorial 2 and first construct a graph meeting the bound obtained there.]

[CO4]

[1+2+2=5 points]

II.2 Let T be a tree.

- Show that the size of the maximum independent set of T is at least $\lceil \frac{|T|}{2} \rceil$.
- Now, let T be a spanning tree for some graph G . Given any $e \in E(T)$, define $\text{cut}(T, e)$ to be the set of edges in $E(G) \setminus E(T)$ whose endpoints lie in the two different components of $T \setminus \{e\}$. Also, given any $f \in E(G) \setminus E(T)$, define the set $\text{cycle}(T, f)$ to be the set of edges in $E(T)$ which form a cycle with f . Then, show that $f \in \text{cut}(T, e)$ iff $e \in \text{cycle}(T, f)$.

[CO2]

[2.5+2.5=5 points]

II.3 Consider a graph G .

- G is k -connected iff for every $T \subseteq S \subseteq V(G)$ with $|T| = 2$ and $|S| = k$, there exists a cycle containing $V(T)$ but bypassing $V(S \setminus T)$.
- Show that $|G| \geq \kappa(G)(\text{diam}(G) - 1) + 2$.

[CO2, CO3, CO4]

[3.5+1.5=5 points]

II.4 Let G be graph with $\lambda(G) = c$. Let $p \in (0, 1)$, and let $G(p)$ be a random graph obtained by independently dropping every edge of G with probability p .

- Show that $\Pr(G(p) \text{ has at least } k \text{ components}) \leq \sum_{\{v_1, \dots, v_k\} \subseteq V(G)} \Pr(v_i \text{ lie in different components of } G(p))$.
- Let $C_1, \dots, C_k$ be k components of $G(p)$. Show that the minimum number of edges whose endpoints do not lie in the same component is at least $\frac{kc}{2}$.
- Show that $\Pr(G(p) \text{ has at least } k \text{ components}) \leq \binom{n}{k} p^{\frac{kc}{2}}$.

[CO1, CO4]

[1.5+1.5+2=5 points]

III. WALL SINA

III.1 A few definitions first. We say that a graph G is *minimally k -connected* if for any $e \in E(G)$, $G \setminus \{e\}$ is no longer k -connected. Given a graph G , a pair of vertices $\{u, v\} \subseteq V(G)$ is said to be *compatible* if for any $u - v$ path P , if $\{w_1, w_2\} \subseteq V(P)$ and $\{w_1, w_2\} \in E(G)$, then the edge $\{w_1, w_2\} \in E(P)$.

- Let G' be graph containing a cycle with a chord. Show that G' is not minimally 2-connected.
- Now, let G be minimally 2-connected, and let $\{u, v\} \subseteq V(G)$ form a pair of compatible vertices. Let P be any $u - v$ path. We shall show that P must contain a degree 2 vertex. To start our proof, assume the contrary. Show that $H = G \setminus E(P)$ does not contain isolated vertices, and every component of H contains at least two vertices from $V(P)$.
- Let $P = x_0x_1 \dots x_k$ where $x_0 = u$ and $x_k = v$. Let $x_i, x_j, i < j$ belong to the same component in H . Then, show that $i \leq j - 2$.
- Let $i \leq j - 2$ such that x_i, x_j lie in the same component of H , and $j - i$ is minimal. Justify that there exists an $l \notin \{i, i + 1, j\}$ such that x_{i+1} and x_l lie in the same component of H .
- Show that $l \notin (i + 1, j)$.
- Show that $l < j$.
- Show that $l > i$.
- Argue that we have a contradiction to the statement in III.1.b) thereby completing the proof.

[CO3,CO4]

[2+1.5+1.5+0.5+0.5+0.5+1.5+1.5+0.5=10 points]

III.2 Recall that a graph G is called a median graph if for any $u, v, w \in V(G)$, there exists a unique $z \in V(G)$, such that any $u_1, u_2 \in \{u, v, w\}$ satisfies $d_G(u_1, z) + d_G(u_2, z) = d_G(u_1, u_2)$. We shall use the notation $m(u, v, w) = z$.

- Let G be a median graph, and let $u, v, w \in V(G)$. Define $I(x, y)$ to be the set of vertices that lie on the shortest $x - y$ paths. Then, show that $\{z\} = |I(u, v) \cap I(v, w) \cap I(u, w)|$, where $z = m(u, v, w)$.
- Let the triple u, v, w have repeated vertices. Show that $m(u, v, w)$ is the vertex that is repeated in the tuple u, v, w .
- Now, let G and H be median graphs, and consider $G \square H$. Show that for any $\{(u_1, u_2), (v_1, v_2)\} \in V(G \square H)$ their distance $d_{G \square H}((u_1, u_2), (v_1, v_2)) = d_G(u_1, v_1) + d_H(u_2, v_2)$.
- Now, consider the triple $(u_1, u_2), (v_1, v_2), (w_1, w_2) \in V(G \square H)$. Define $z := (z_1, z_2) = (m(u_1, v_1, w_1), m(u_2, v_2, w_2))$. Show that z satisfies the properties of the median, i.e., for any $(x_1, x_2), (y_1, y_2) \in \{(u_1, u_2), (v_1, v_2), (w_1, w_2)\}$, we have $d_{G \square H}((x_1, x_2), z) + d_{G \square H}((y_1, y_2), z) = d_{G \square H}((x_1, x_2), (y_1, y_2))$.
- Show that z is indeed the median (i.e., show its uniqueness), thereby concluding that box product of median graphs is a median graph.

[Hint: You may use parts a) and c)]

[CO2,CO4]

[1.5+1+2.5+2+3=10 points]